

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: 11 to 12 kB expression vector, Rep 4.
Species Used	Hamster, CHO -K1, ovary, (requires proline)		

Before the Pulse

Cell Growth Medium	F12 + 10% serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	10 min., on ice.
Wash Solution	Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	Room temperature (pulse), then ice	Cuvette Gap	0.4 cm
Electroporation Medium*	Phosphate Buffered Saline	Voltage	0.20 to 0.40 kV
Cell Density	2 x 10 ⁵ (5) cells / pulse	Field Strength	0.5 to 1.0 kV/cm
Volume of Cells	0.8 ml	Capacitor	500 & 960 µF
DNA Concentration	0.5 µg / pulse	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Not given	Time Constant	18 to 22 msec
Volume of DNA	Not given		
After the Pulse			
Outgrowth Medium	F12 + serum + hygromycin or neomycin		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	14 to 21 days
Selection Method or Assay Used	Hygromycin or neomycin selection
Electroporation Efficiency	Not done as yet - just beginning
Per Cent Survival	>10%

Name of Submitter Not listed

Institution Address

Survey Number

086